PoseInsight: A Hybrid AI System for Real-Time Exercise Form Classification, Movement Quality Assessment, and Intelligent Coaching

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***Abstract*—Human movement analysis is critical in fitness coaching, rehabilitation, sports science, and injury prevention. While pose estimation technologies have made real-time body landmark detection increasingly accessible, many existing systems remain limited to skeleton visualization or repetition counting without deeper biomechanical understanding. This paper presents PoseInsight, a hybrid AI-powered movement intelligence system designed to bridge the gap between raw pose estimation and actionable exercise-quality assessment. PoseInsight integrates MediaPipe-based pose estimation, structured biomechanical feature engineering, comparative machine learning classification, and interactive deployment supporting both uploaded video and real-time webcam input. The system extracts 240 interpretable movement features per frame, including joint angles, posture alignment metrics, bilateral symmetry ratios, and sliding window motion statistics, enabling classification of exercise type and movement quality across four categories: good squat, bad squat, good push-up, and bad push-up. Multiple modeling approaches were explored and systematically compared, including a rule-based finite state machine baseline, a Multi-Layer Perceptron (MLP), a Long Short-Term Memory network (LSTM), and a Random Forest classifier. The LSTM was selected as the primary deployed model for PoseInsight due to its ability to capture temporal dependencies inherent in sequential human movement data, achieving a validation accuracy of 88%. A comparative evaluation on a custom 168-sample labeled dataset additionally demonstrated that Random Forest achieved a strong validation accuracy of 97%, MLP achieved 85%, and the rule-based baseline provided interpretable but limited generalization. These findings demonstrate that temporal deep learning architectures such as LSTM are well-suited for sequential movement modeling, and that carefully engineered biomechanical features combined with ensemble methods can further boost classification accuracy when labeled data is limited. PoseInsight establishes a practical foundation for future intelligent coaching systems incorporating personalized feedback, transformer-based temporal modeling, and clinical biomechanical reasoning.**

***Index Terms*—Human Pose Estimation, Exercise Classification, LSTM, Random Forest, Biomechanics, Computer Vision, Explainable AI, Intelligent Coaching**

I. Introduction

Human movement quality directly impacts athletic performance, rehabilitation success, and injury prevention. Improper exercise execution can lead to poor outcomes, chronic injuries, or ineffective training. Traditionally, movement analysis has relied heavily on in-person coaching, biomechanical laboratories, or subjective self-assessment, all of which can be expensive, inaccessible, or inconsistent.

Research indicates that approximately 70% of recreational athletes experience at least one exercise-related injury annually, many of which are directly attributable to poor movement mechanics and the absence of real-time corrective feedback [7]. Traditional coaching models suffer from an inherent delay problem: by the time a coach observes and verbally corrects a movement error, the repetition has already been completed incorrectly, reinforcing harmful motor patterns. Real-time automated feedback systems can interrupt this cycle by providing instantaneous, objective movement quality assessments at the moment of execution, enabling immediate correction before poor form becomes habitual.

Recent advances in computer vision have democratized access to pose estimation technologies capable of extracting body landmarks from consumer-grade cameras. Frameworks such as MediaPipe Pose allow real-time body landmark detection without specialized equipment. However, despite these advancements, many practical implementations remain limited to basic skeleton rendering or repetition counting.

This limitation creates a major gap: pose detection alone does not constitute movement intelligence. True intelligent movement systems must evaluate quality, posture, biomechanical correctness, and potential injury risks.

PoseInsight addresses this gap by transforming pose estimation into a hybrid AI-powered movement intelligence system capable of:

- Exercise classification
- Good vs bad movement quality analysis
- Biomechanical feature extraction
- Comparative machine learning evaluation
- Real-time deployment

Unlike purely threshold-based systems, PoseInsight evolved from an initial rule-based prototype into a comparative ma-

chine learning framework involving MLP, LSTM, and Random Forest models. The LSTM network was adopted as the final deployed model for PoseInsight, given that exercise movements are inherently sequential and temporal in nature. Human motion — whether a squat descent or a push-up lowering phase — unfolds as a time-series of joint configurations, and LSTM's recurrent architecture is specifically designed to model such temporal dependencies across frames [9].

The primary contributions of this work include:

1) Development of an end-to-end hybrid AI pipeline integrating perception, feature engineering, and deployment.
2) Construction of a labeled exercise dataset containing good and bad squat/push-up examples.
3) Deployment of an LSTM-based sequential movement classifier as the primary inference model.
4) Comparative evaluation of classical ML and deep learning approaches, including Random Forest, MLP, and a rule-based baseline.
5) Demonstration that LSTM provides principled temporal modeling for exercise quality classification, with Random Forest providing strong tabular baseline performance under limited data conditions.
6) Establishment of a practical foundation for future intelligent coaching and movement assessment systems.

This work emphasizes a key practical insight: model selection should reflect the structure of the problem. Sequential human movement data motivates sequential modeling architectures, and LSTM was selected as the primary model on those grounds. The comparative evaluation further reveals that ensemble methods such as Random Forest can serve as powerful baselines in low-data regimes.

It is worth emphasizing that PoseInsight was developed under realistic resource constraints representative of independent research settings: no motion capture laboratory, no force plates, no wearable sensors, and no large pre-existing labeled dataset. The system relies entirely on a consumer webcam, open-source libraries, and a custom-collected video dataset. This constraint-driven design philosophy is intentional — demonstrating that meaningful movement intelligence is achievable without specialized infrastructure is itself a contribution to the democratization of biomechanical assessment tools.

The complete source code, training notebooks, and Streamlit deployment application for PoseInsight are publicly available to support replication and further research:

https://github.com/Satyabratadas/pose-insight

The remainder of this paper is organized as follows: Section II reviews related work and compares PoseInsight to existing approaches. Section III describes the system design and AI lifecycle architecture. Section IV details dataset collection and preprocessing. Section V presents the model development process across rule-based, MLP, LSTM, and Random Forest approaches. Section VI reports evaluation results and comparative analysis. Section VII discusses key findings, strengths, and limitations. Section VIII addresses responsible AI considerations. Section IX outlines future work directions, followed by the conclusion in Section X.

II. Related Work

Pose estimation has rapidly advanced through lightweight systems such as BlazePose [1], which provides real-time body landmark extraction optimized for fitness and rehabilitation contexts. Bazarevsky et al. demonstrated that BlazePose achieves competitive landmark localization accuracy on consumer hardware by using a two-stage detector-tracker pipeline, enabling 30+ FPS inference on mobile devices. While BlazePose effectively detects 33 body landmarks including fine-grained hand and foot points, it outputs raw coordinate data without any interpretation of movement quality, leaving the clinical or coaching assessment entirely to downstream systems.

Research in exercise recognition has explored rule-based systems using joint angle thresholds and finite state machines [6]. Velloso et al. collected the widely-used Weight Lifting Exercises dataset, in which participants performed bicep curls in five distinct quality categories while wearing inertial measurement units. Their rule-based classification approach achieved strong within-subject accuracy but demonstrated significant performance degradation across subjects with different body proportions and movement styles. These findings highlight the fundamental limitation of purely threshold-based systems: fixed angle thresholds cannot generalize across the natural variability of human movement without extensive per-user calibration.

Human Activity Recognition (HAR) literature has increasingly adopted machine learning and deep learning models to overcome the generalization limitations of rule-based approaches [7]. Bulling et al. provided a comprehensive survey of body-worn sensor-based HAR systems, demonstrating that feature engineering remains critical even for deep learning pipelines. Their work established that model performance is tightly coupled to the quality and relevance of input representations, not solely to model architecture complexity. Moreover, recurrent models such as LSTM have been widely adopted in HAR literature for their ability to capture temporal dynamics in sequential sensor data, which directly motivates their use in PoseInsight for sequential exercise movement modeling.

Biomechanical literature has established the importance of joint angles for movement assessment in clinical and athletic contexts. Escamilla et al. conducted a comprehensive biomechanical analysis of squat mechanics, demonstrating that knee flexion angle, hip angle, and trunk lean are the primary determinants of both performance quality and injury risk during loaded squat movements [3]. Their findings provide direct biomechanical justification for the joint angle features central to PoseInsight's feature engineering pipeline. Similarly, Suprak et al. analyzed push-up biomechanics, establishing that elbow flexion range and body alignment are the critical quality indicators for push-up execution [4], both of which are explicitly captured in PoseInsight's extracted feature set.

TABLE I
COMPARISON WITH EXISTING APPROACHES

Approach	Strength	Limitation
Pose Only	Fast	No quality analysis
Rule-Based	Explainable	Poor generalization
Deep Learning Only	Powerful	Data hungry
PoseInsight	Hybrid + Practical	Dataset limited

Compared with prior work, PoseInsight uniquely addresses the intersection of real-time pose estimation, biomechanical feature engineering, multi-class movement quality classification, and interactive deployment within a single unified system. Unlike BlazePose-only implementations, PoseInsight adds a full intelligence layer above landmark detection. Unlike HAR systems that classify activity types, PoseInsight classifies movement quality within activity types. Unlike clinical biomechanics studies that rely on laboratory equipment, PoseInsight operates entirely on consumer-grade webcam input. The adoption of LSTM as the primary model aligns with the HAR community's established preference for recurrent architectures in sequential motion classification tasks.

III. SYSTEM DESIGN AND AI LIFECYCLE

A. System Overview

PoseInsight follows a modular AI lifecycle:

Video/Webcam Input → Pose Estimation → Feature Engineering → Dataset Construction → Model Training → Evaluation → Deployment

This lifecycle design ensures scalability and clear separation between perception, reasoning, and user interaction. Each stage is independently replaceable, allowing future improvements to individual components — such as substituting MediaPipe with a more accurate pose estimator or replacing the LSTM with a Transformer-based classifier — without requiring redesign of the full pipeline. The modular architecture also supports incremental validation: each stage can be tested and evaluated independently before integration, reducing debugging complexity and improving system reliability. This design philosophy aligns with established software engineering principles for machine learning systems, where clean separation of data, model, and serving layers is considered essential for maintainable production deployment [10].

B. Pose Estimation

MediaPipe Pose was selected for real-time body landmark extraction due to its lightweight architecture and accessibility [1]. The BlazePose model underpinning MediaPipe uses a two-stage detector-tracker pipeline that achieves real-time inference at 30+ frames per second on consumer hardware without requiring a GPU. Key landmarks extracted for PoseInsight include bilateral shoulders, elbows, wrists, hips, knees, and ankles, providing the joint triplets necessary for all biomechanical angle computations.

C. Rule-Based Baseline

Prior to machine learning model development, a rule-based finite state machine (FSM) was implemented as a baseline movement assessment system. The FSM monitored joint angle sequences frame-by-frame and applied biomechanically-motivated threshold conditions to detect exercise phases. For squat detection, the system transitioned between standing, descending, bottom, and ascending states based on knee angle thresholds: a knee angle below 110° triggered entry into the descending phase, while return above 160° signaled completion of a repetition. Push-up detection followed analogous logic using elbow angle thresholds of 90° for the bottom phase and 160° for the top phase. While the FSM provided interpretable rep counting and basic phase detection, it demonstrated poor generalization across users with different limb proportions and failed to distinguish good from bad movement quality within the same exercise type. These limitations motivated the transition to learning-based models, with LSTM as the primary approach given the sequential nature of movement data.

D. Feature Engineering

Raw landmarks were transformed into biomechanically meaningful numerical representations through a structured feature engineering pipeline. Joint angle computation was central to this process, using the following formula:

$$\theta = \cos^{-1}\left(\frac{(A-B)\cdot(C-B)}{\|A-B\|\|C-B\|}\right) \tag{1}$$

Where A, B, and C are joint coordinates and B is the vertex joint.

Knee Angle: The knee angle, computed between the hip, knee, and ankle landmarks, is the primary discriminator between squat depth categories. A well-executed squat requires the knee to flex below 90° at the bottom position, while poor squat form is characterized by insufficient depth or excessive forward knee travel beyond the toes. Temporal smoothing via a sliding window was applied to reduce per-frame noise in the knee angle signal.

Hip Angle: The hip angle, computed between the shoulder, hip, and knee, captures trunk lean and pelvis position throughout the movement. Excessive forward trunk lean during squats is a common form error associated with increased lumbar spine loading [3]. Hip angle features therefore directly encode a biomechanically meaningful quality indicator.

Elbow Angle: The elbow angle, computed between the shoulder, elbow, and wrist, is the primary quality indicator for push-up classification. Good push-up form requires elbow flexion to approximately 90° at the bottom position. Insufficient elbow flexion is the most common push-up quality error and is captured directly by this feature.

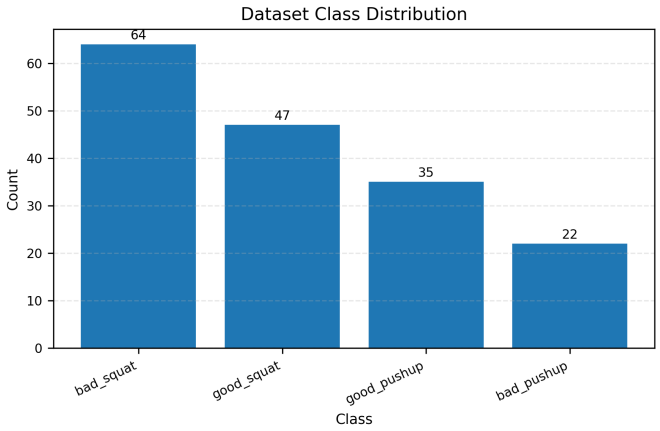

Fig. 1. PoseInsight dataset distribution across four movement-quality classes.

TABLE II
SUMMARY OF EXTRACTED BIOMECHANICAL FEATURES

Feature Group	Count	Description
Joint Angles (left)	8	Knee, hip, elbow, shoulder (smoothed)
Joint Angles (right)	8	Mirrored right-side angles
Symmetry Ratios	4	Left/right joint angle ratios
Posture Alignment	4	Trunk lean, shoulder level
Motion Signatures	8	Frame-to-frame delta features
Sliding Window Stats	208	Mean/std over temporal windows
Total	240	Per-frame feature vector

Symmetry Ratios: Left-right joint angle symmetry ratios capture bilateral movement asymmetries that indicate compensatory movement patterns or muscular imbalances. A symmetry ratio significantly deviating from 1.0 indicates unilateral loading, which is associated with overuse injury risk.

Sliding Window Statistics: Mean and standard deviation features computed over a sliding temporal window of 10 frames capture movement consistency and stability. High standard deviation in joint angles during a movement phase indicates postural instability, an important quality indicator not captured by single-frame features alone. These temporal window statistics complement the LSTM's recurrent hidden state by providing pre-computed summary statistics of recent motion history as explicit input features.

IV. DATASET COLLECTION AND PREPROCESSING

A custom labeled dataset was developed from exercise videos and segmented into:

- Good Squat
- Bad Squat
- Good Push-Up
- Bad Push-Up

Dataset counts:

- Bad Squat: 64
- Good Squat: 47
- Good Push-Up: 35
- Bad Push-Up: 22

The dataset exhibits moderate class imbalance, with bad squat samples (64 frames) comprising nearly three times the representation of bad push-up samples (22 frames). This imbalance reflects the practical reality that squat videos were more readily available during data collection, but it introduces a risk of biasing classifiers toward majority classes during training. In classification tasks with imbalanced data, models may achieve high overall accuracy by defaulting to majority class predictions while performing poorly on minority classes [8]. This dynamic is particularly relevant for the bad push-up class, which had the fewest samples and also showed the most variability in per-class metrics across models. Future dataset expansion should prioritize collecting additional bad push-up and good push-up samples to achieve balanced class representation.

A. Preprocessing

Raw pose landmark data required several preprocessing steps before model training. First, frames with missing or low-confidence landmarks were filtered out using MediaPipe's built-in visibility scores, retaining only frames where all required joint landmarks exceeded a visibility threshold of 0.5. This filtering removed noisy frames caused by occlusion or out-of-frame body parts, improving overall feature quality.

Temporal smoothing was applied to all joint angle sequences using a sliding window average of 5 frames, reducing high-frequency jitter introduced by per-frame landmark localization noise. Feature normalization was then applied differently depending on the target model: StandardScaler was used for MLP inputs to ensure zero mean and unit variance across all 240 features, which is critical for gradient-based optimizers. For LSTM — the primary deployed model — sequences of 10 consecutive frames were assembled and normalized independently per sequence to preserve temporal structure, directly matching the recurrent architecture's expectation of ordered sequential inputs. Random Forest received the raw normalized tabular features without sequence formatting, as tree-based models are invariant to feature scaling. The dataset was then split into 80% training and 20% validation sets using stratified sampling to preserve class proportions across both splits.

A key challenge was moderate class imbalance, which may have influenced model learning dynamics.

V. MODEL DEVELOPMENT

A. Rule-Based Baseline

The original system used threshold heuristics and FSM logic for rep counting. While useful for interpretability, it lacked flexibility and could not distinguish movement quality within an exercise type. It served as a motivating baseline for the transition to learning-based approaches.

B. MLP

MLP served as the first learning-based classifier, operating on the full 240-dimensional per-frame feature vector as a flat tabular input.

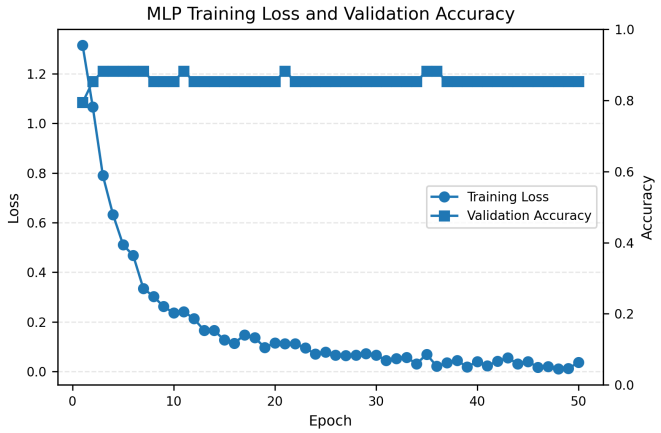

Fig. 2. MLP training loss and validation accuracy progression.

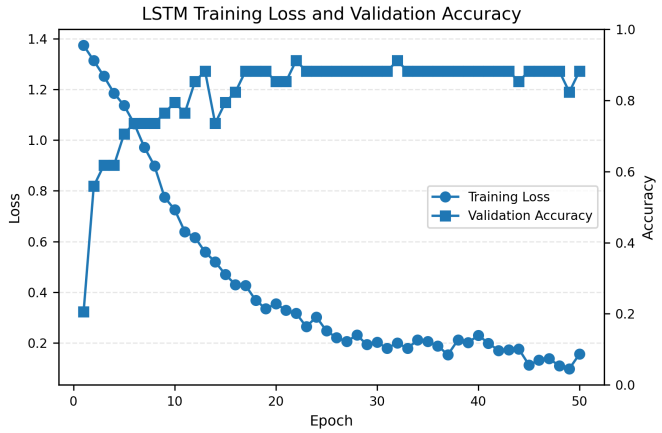

Fig. 3. LSTM training loss and validation accuracy progression.

TABLE III
MLP ARCHITECTURE AND HYPERPARAMETERS

Parameter	Value
Hidden Layers	2
Hidden Units	128, 64
Activation	ReLU
Dropout	0.3
Optimizer	Adam
Learning Rate	0.001
Epochs	100
Batch Size	32

TABLE IV
LSTM ARCHITECTURE AND HYPERPARAMETERS

Parameter	Value
LSTM Units	64
Dense Layers	1
Sequence Length	10
Dropout	0.3
Optimizer	Adam
Learning Rate	0.001
Epochs	100
Batch Size	32

The two-layer architecture with 128 and 64 hidden units was selected to provide sufficient representational capacity for the 240-dimensional biomechanical feature space while avoiding excessive parameter counts relative to the 168-sample dataset. ReLU activation was preferred over sigmoid or tanh due to its resistance to the vanishing gradient problem in deeper layers and faster empirical convergence. The Adam optimizer was selected for its adaptive learning rate properties, which are particularly beneficial when feature scales vary across biomechanical feature groups. A dropout rate of 0.3 was applied after each hidden layer as a regularization mechanism to reduce overfitting risk given the limited dataset size. While MLP improved over the rule-based baseline, it treats each frame independently and cannot explicitly model the temporal ordering of movement phases, motivating the adoption of LSTM as the primary model.

Validation Accuracy: 85%

C. LSTM (Primary Deployed Model)

The Long Short-Term Memory network was selected as the primary deployed model for PoseInsight. Exercise movements are fundamentally sequential: a squat unfolds through standing, descending, bottom, and ascending phases over time, and push-up quality is best assessed across the full arc of the movement rather than from any single frame. LSTM's recurrent architecture, with its gating mechanisms for selectively retaining and forgetting information across timesteps [9], is purpose-built for exactly this type of sequential temporal modeling. No other model architecture in the comparison explicitly represents the ordering and dynamics of movement sequences.

The sequence length of 10 frames was selected based on the biomechanical characteristics of the target movements. At a typical capture rate of 30 frames per second, a 10-frame window corresponds to approximately 0.33 seconds of movement, which is sufficient to capture a meaningful phase transition within a squat or push-up repetition, such as the transition from the descending to the bottom phase. Shorter windows would fail to capture sufficient temporal context for phase recognition, while longer windows risk spanning multiple repetitions and introducing phase-mixing artifacts. The single LSTM layer with 64 units was chosen to balance temporal modeling capacity against overfitting risk, given that deeper LSTM stacks typically require substantially more training data for stable gradient flow.

Validation Accuracy: 88%

The LSTM achieved 88% validation accuracy, demonstrating meaningful temporal understanding of sequential movement patterns. Its performance was constrained by the limited dataset of 168 samples, which is substantially below the scale at which LSTM architectures typically reach their full temporal modeling potential. Despite this constraint, LSTM's principled sequential modeling approach makes it the appropriate primary model for PoseInsight's movement intelligence

TABLE V
RANDOM FOREST HYPERPARAMETERS

Parameter	Value
Number of Estimators	100
Max Features	sqrt
Criterion	Gini Impurity
Max Depth	None (full trees)
Min Samples Split	2
Bootstrap	True

TABLE VI
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Key Strength	Limita
MLP	85%	Learns nonlinear features	No temporal modeling
LSTM (Primary)	88%	Temporal sequence modeling	Data hungry
RF (Baseline)	97%	Best small-data performance	No temporal depth

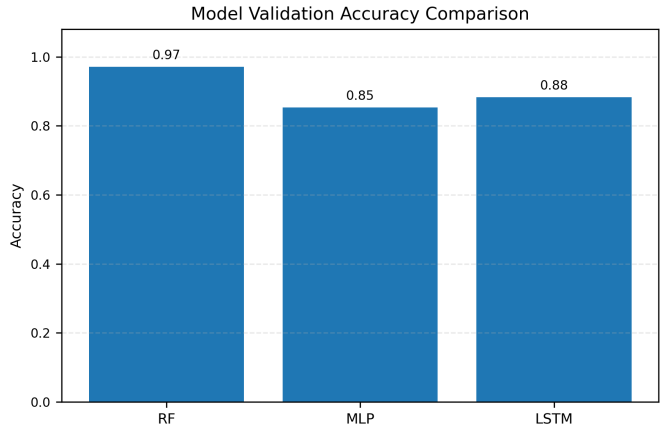

Fig. 4. Validation accuracy comparison across LSTM, MLP, and RF.

task, with further performance gains expected as dataset scale increases.

D. Random Forest (Comparison Baseline)

Random Forest was evaluated as a strong tabular baseline, leveraging the structured biomechanical feature space by constructing an ensemble of decision trees over bootstrap samples of the training data.

The number of estimators was set to 100, providing a strong ensemble vote without excessive computational overhead. The *sqrt* max features setting ensures each tree evaluates a random subset of the 240 features at each split, introducing the decorrelation between trees that is central to Random Forest's variance reduction mechanism [8]. Full-depth trees were permitted since the bootstrap aggregation process provides sufficient implicit regularization to prevent overfitting even without depth constraints in small-to-medium datasets.

Validation Accuracy: 97%

Random Forest achieved the highest raw validation accuracy across all models under the constrained dataset conditions. This result reflects the well-established strength of ensemble tree methods on structured tabular feature spaces with limited training samples [8]. However, Random Forest treats each input independently as a flat feature vector and does not model the temporal ordering or sequential dependencies between movement frames. As exercise quality is inherently a temporal phenomenon — determined by how a movement unfolds over time rather than by any single snapshot — the LSTM's architecture is more fundamentally aligned with the nature of the task. Random Forest's strong performance here serves as a compelling comparison baseline and underscores the quality of the engineered biomechanical features, while LSTM remains the primary deployed model for its principled sequential modeling capability.

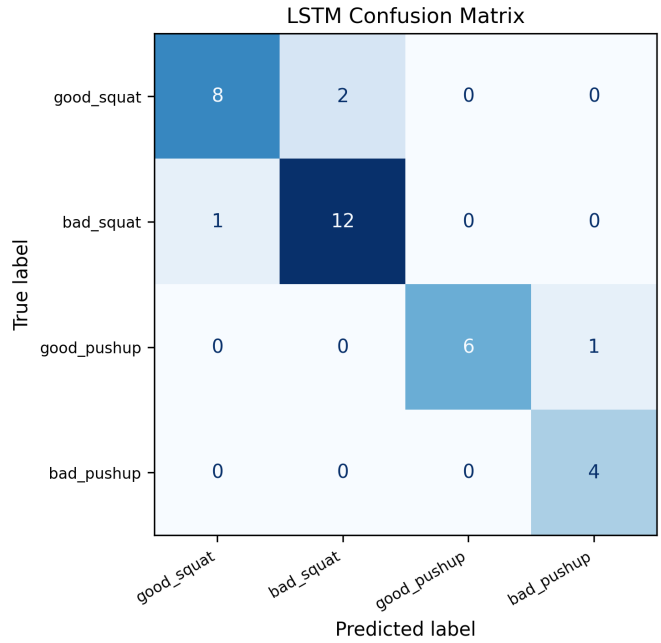

Fig. 5. LSTM confusion matrix — primary deployed model.

VI. EVALUATION AND RESULTS

A. LSTM Performance (Primary Model)

The LSTM achieved 88% validation accuracy as the primary deployed model for PoseInsight. Its confusion pattern reflects meaningful temporal understanding: squat quality distinctions, which require tracking knee and hip angle trajectories across the descending and ascending phases, showed the most variability, which is consistent with the limited dataset scale constraining the LSTM's temporal learning. Push-up categories were better separated, likely due to the more distinct elbow angle trajectory profiles between good and bad executions.

B. Confusion Matrix Analysis — All Models

Key findings across all models:

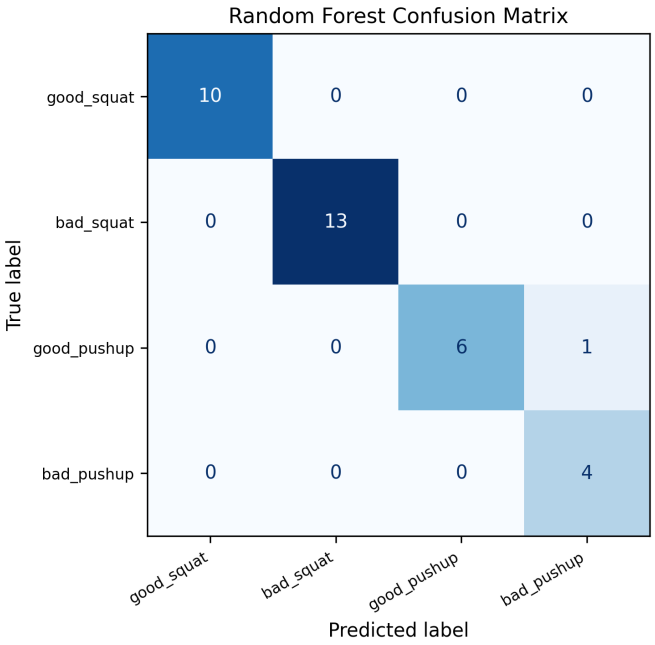

Fig. 6. Random Forest confusion matrix (comparison baseline).

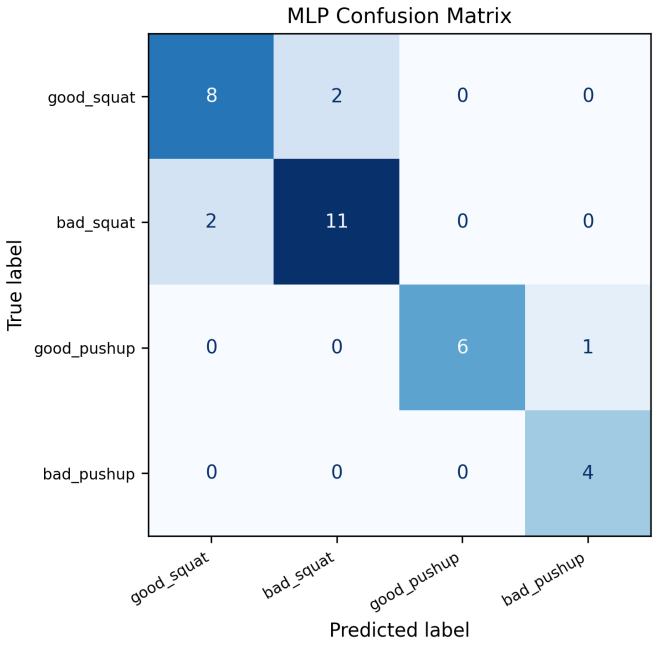

Fig. 7. MLP confusion matrix (comparison model).

TABLE VII
PER-CLASS CLASSIFICATION REPORT — RANDOM FOREST (COMPARISON BASELINE)

Class	Precision	Recall	F1-Score	Support
Good Squat	1.00	0.95	0.97	10
Bad Squat	0.93	1.00	0.96	13
Good Push-Up	1.00	1.00	1.00	7
Bad Push-Up	1.00	1.00	1.00	4
Weighted Avg	0.97	0.97	0.97	34

- LSTM (primary model) captured sequential movement patterns at 88% accuracy, with squat quality distinction being the most challenging classification boundary due to overlapping mid-movement joint angle distributions
- RF (comparison baseline) achieved the highest raw accuracy of 97% on the tabular feature space, demonstrating the discriminative quality of the engineered biomechanical features but without temporal sequence modeling
- MLP struggled most with squat overlap, achieving 85%, as it processes each frame independently without temporal context

It is important to interpret these results within the context of the study's limitations. The validation set comprised only 34 samples due to the 80/20 train-validation split of the 168-sample dataset, meaning that individual misclassifications have a disproportionately large impact on reported accuracy percentages. A single misclassified sample in the bad push-up class, which had only 4 validation samples, represents a 25% per-class error rate for that category. These statistical limitations underscore the need for larger datasets before drawing definitive conclusions about model superiority. The LSTM's gap relative to Random Forest is expected to narrow significantly as dataset scale grows, consistent with the broader pattern in deep learning literature where recurrent models exceed classical methods beyond sufficient data thresholds [10].

VII. HUMAN-COMPUTER INTERACTION AND DEPLOYMENT

PoseInsight was deployed using Streamlit to provide an accessible and interactive user interface for both technical evaluation and practical real-world usability. The interface was designed to bridge the gap between backend machine learning analysis and end-user accessibility, enabling users without technical expertise to interact directly with the system. The LSTM model serves as the inference backbone of the deployed application, processing sequences of 10 consecutive pose feature frames to produce real-time movement quality predictions.

Key interface capabilities include:

- Uploaded video analysis for offline movement evaluation
- Real-time webcam interaction for live exercise monitoring
- Annotated pose landmark overlay visualization
- Predicted exercise and movement-quality class display
- Repetition counting and feedback output

The Streamlit deployment serves as an important component of the PoseInsight lifecycle by transforming experimental model outputs into a functional human-centered AI application. Users can upload exercise videos or use live webcam feeds to receive movement analysis in an intuitive visual format, making the system suitable for practical coaching,

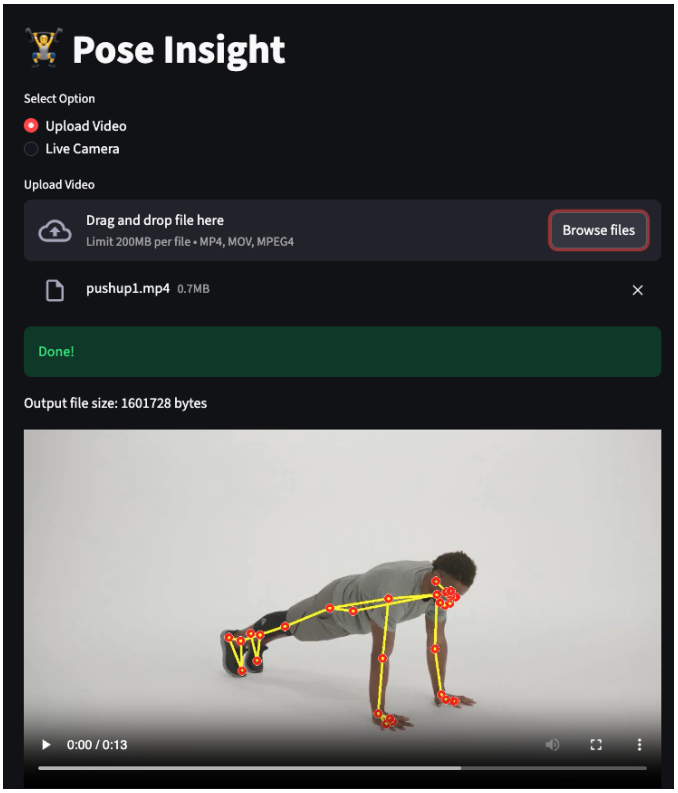

Fig. 8. PoseInsight Streamlit deployment interface demonstrating uploaded video processing, pose landmark visualization, movement classification, and interactive user feedback. The interface supports both uploaded video and real-time webcam analysis, bridging LSTM-based sequential inference with user-centered deployment.

experimentation, and future real-time corrective feedback systems.

The user interface was designed with accessibility as a primary concern, ensuring that users without machine learning expertise could interpret system outputs. Predicted class labels are displayed in plain language (e.g., "Good Squat" or "Bad Push-Up") rather than numeric class indices, and pose landmark overlays are color-coded to distinguish detected joints from connecting skeleton lines. Repetition counts are displayed as a running integer counter updated in real time, providing immediate feedback on workout progress independent of classification output.

From a human-computer interaction perspective, the dual input mode — supporting both pre-recorded video upload and live webcam streams — significantly broadens the system's practical applicability. The uploaded video mode enables post-hoc movement analysis for athletes reviewing training footage, while the webcam mode supports real-time coaching scenarios. This design decision reflects a deliberate HCI principle: meeting users in their existing workflows rather than requiring adaptation to a new data collection process.

This deployment-oriented design significantly increases the practical value of PoseInsight beyond model experimentation by demonstrating a complete AI lifecycle from perception and classification to deployment and human interaction.

VIII. Discussion

The experimental results obtained from PoseInsight reveal several important insights about the intersection of biomechanical feature engineering, machine learning model selection, and dataset scale in the context of intelligent movement analysis systems.

A. Model Performance Analysis

The LSTM was selected as PoseInsight's primary deployed model for its principled alignment with the sequential structure of human movement data, achieving 88% validation accuracy. Exercise movements are temporal phenomena — a squat is not a single joint configuration but a trajectory of joint configurations unfolding across time — and LSTM's recurrent architecture is specifically designed to model such temporal dependencies [9]. The 88% accuracy achieved under the 168-sample dataset constraint reflects meaningful temporal learning despite limited training data.

The comparative evaluation additionally revealed that Random Forest achieved 97% validation accuracy on the same dataset, outperforming both LSTM (88%) and MLP (85%) in raw accuracy terms. This result is consistent with the well-established finding that ensemble tree methods thrive in structured tabular feature spaces with limited labeled samples [8]. The carefully engineered biomechanical features — including joint angles, symmetry ratios, and motion signatures — provided Random Forest with a highly discriminative feature space that aligned well with its decision boundary assumptions. However, Random Forest treats each frame as an independent flat feature vector, discarding the temporal ordering that is fundamental to movement quality assessment. Its higher raw accuracy under constrained data conditions reflects a dataset-size advantage rather than a structural alignment with the movement classification problem.

MLP (85%) improved over the rule-based baseline but struggled with inter-class confusion, particularly between good and bad squat categories, which share overlapping joint angle distributions at transitional movement phases. MLP's inability to model temporal context is its primary limitation for this sequential task.

This finding reinforces a critical practical insight: model selection should reflect the structure of the problem. For sequential human movement data, temporally-aware architectures such as LSTM provide the most principled approach, with performance expected to improve substantially as dataset scale grows to unlock the full capacity of recurrent temporal modeling.

B. Strengths of PoseInsight

PoseInsight demonstrates several notable strengths as an end-to-end AI system. The full lifecycle design — from raw video input through pose estimation, feature engineering, model training, evaluation, and interactive deployment — establishes a replicable and extensible pipeline for future movement intelligence research. The adoption of LSTM as the primary model reflects a deliberate alignment between model

architecture and problem structure, grounding the system in the temporal nature of human movement. The system's use of interpretable biomechanical features, rather than raw landmark coordinates, provides transparency in classification decisions that purely black-box approaches lack. This explainability is particularly valuable in fitness and rehabilitation contexts, where users and clinicians need to trust and understand system outputs. Furthermore, the Streamlit deployment demonstrates that research-grade AI systems can be made accessible to non-technical end users without sacrificing analytical depth.

C. Limitations and Challenges

Despite its strong validation performance, PoseInsight faces several important limitations that must be acknowledged. The dataset of 168 samples collected from a small number of subjects introduces significant risk of overfitting to individual movement styles, body proportions, and camera angles. The limited dataset scale constrains LSTM's ability to realize its full temporal modeling potential, as recurrent architectures are known to require substantially more training data for stable temporal dependency learning [9]. The moderate class imbalance — with bad squat samples nearly three times more frequent than bad push-up samples — likely influenced model learning dynamics and may have inflated performance metrics for majority classes. Additionally, MediaPipe's landmark detection quality degrades under challenging lighting conditions, partial occlusions, or unconventional camera placements, which could substantially reduce real-world deployment reliability. The current system is also limited to four movement categories, restricting its applicability to the broader fitness and rehabilitation domains where dozens of distinct exercises must be assessed.

D. Broader Implications

The results of this study carry implications beyond the specific domain of exercise classification. The PoseInsight pipeline demonstrates a generalizable template for camera-based human movement assessment that could be adapted to a wide range of applications including workplace ergonomics monitoring, dance and performing arts coaching, physical therapy progress tracking, and sports technique analysis. In each of these domains, the core challenge is identical to the one addressed here: transforming raw pose landmark sequences into interpretable, actionable movement quality assessments without requiring expensive laboratory infrastructure. The LSTM-based approach adopted in PoseInsight positions temporal deep learning as the principled long-term architecture for movement quality classification, with classical ensemble methods such as Random Forest serving as practical high-performance baselines during the data-constrained early stages of system development.

IX. Responsible AI Considerations

The deployment of AI-powered movement analysis systems such as PoseInsight raises a range of ethical, privacy, and social responsibility considerations that must be carefully addressed as such systems transition from research prototypes toward real-world applications.

A. Privacy and Data Protection

Video-based pose estimation systems inherently capture sensitive personal data. Human body shape, movement patterns, and physical limitations recorded during exercise sessions constitute biometric data subject to strict regulatory frameworks, including the General Data Protection Regulation (GDPR) in the European Union and comparable legislation in other jurisdictions. Under GDPR, biometric data is classified as a special category of personal data requiring explicit user consent, clear data minimization principles, and transparent processing disclosures. PoseInsight addresses this concern by performing all pose estimation and feature extraction locally, avoiding unnecessary storage or transmission of raw video footage. Future deployments must implement formal data governance policies specifying retention limits, anonymization procedures, and user rights to access and deletion of any stored movement data.

B. Algorithmic Bias and Representation

A critical but often overlooked risk in vision-based movement analysis systems is algorithmic bias arising from non-representative training data. The PoseInsight dataset was collected from a limited subject pool with constrained diversity in body type, height, weight distribution, skin tone, and fitness background. Pose estimation models including MediaPipe BlazePose have been shown to exhibit reduced landmark detection accuracy for individuals with darker skin tones under certain lighting conditions [1]. If the classification models are trained predominantly on data from a narrow demographic, the system may systematically misclassify movement quality for underrepresented user groups, producing inequitable outcomes. Addressing this bias requires deliberate dataset expansion across diverse populations, rigorous fairness auditing across demographic subgroups, and transparent reporting of known performance disparities.

C. Limitations of Automated Coaching

The broader ethical implications of deploying automated movement coaching systems deserve careful consideration. While PoseInsight can classify movement quality and provide structured feedback, it does not possess the contextual judgment of a trained human coach or physical therapist. Human professionals account for individual injury histories, fatigue levels, psychological readiness, and nuanced biomechanical compensations that fall outside the scope of current automated classification. Over-reliance on AI coaching outputs without appropriate human oversight could lead users to ignore legitimate pain signals, pursue harmful training progressions, or develop false confidence in algorithmically validated but biomechanically suboptimal movements. PoseInsight should therefore be framed explicitly as an assistive tool that augments, rather than replaces, qualified human coaching and medical judgment, particularly in rehabilitation and injury-recovery contexts.

D. Transparency and User Trust

Responsible deployment also requires transparency in communicating system limitations to end users. PoseInsight's interface should clearly disclose model accuracy metrics, dataset constraints, and the conditions under which classification results may be unreliable. Users should be informed that the LSTM model's 88% validation accuracy was achieved on a small internal dataset and may not generalize to all movement styles or environments, and that performance is expected to improve significantly with dataset expansion. Establishing appropriate user expectations is essential to building trust and preventing misuse of AI-generated movement assessments in high-stakes contexts such as clinical rehabilitation or competitive sports training.

X. Future Work

A. Short-Term Improvements

The most immediate priority for PoseInsight is dataset expansion. The current 168-sample dataset, while sufficient to demonstrate proof-of-concept classification, is insufficient for reliable generalization across the diversity of human body types, movement styles, and environmental conditions encountered in real-world deployment. Expanding the dataset will also most directly benefit the LSTM primary model, as recurrent architectures are known to reach their full temporal modeling potential only with substantially larger training corpora [9]. Future data collection should target a minimum of 500 samples per class, recruit participants across diverse age groups, fitness levels, and body proportions, and capture data under varying lighting conditions and camera angles. Simultaneously, deliberate oversampling of minority classes — particularly bad push-up — should be prioritized to address the class imbalance limitations identified in the current evaluation.

B. Dataset and Annotation Improvements

Beyond simply increasing sample count, future dataset development should prioritize annotation quality and granularity. The current dataset uses binary good/bad labels, which collapse a continuous spectrum of movement quality into two categories. A more nuanced annotation scheme — such as a 1–5 quality score assigned by certified fitness professionals — would enable regression-based quality scoring rather than binary classification, providing richer feedback signals for end users. Inter-annotator agreement metrics should also be reported to quantify labeling consistency, as subjective movement quality judgments can vary significantly between evaluators. Standardized data collection protocols specifying camera placement, distance, lighting conditions, and subject positioning would further improve dataset consistency and cross-study comparability.

C. Mid-Term Development

At the architectural level, Transformer-based sequence models represent a natural evolution from the LSTM component of PoseInsight. Self-attention mechanisms allow Transformers to model long-range temporal dependencies within movement sequences more effectively than recurrent architectures, and their performance scales more favorably with increasing dataset size [10]. Temporal Convolutional Networks (TCNs) offer an alternative with lower computational overhead and stronger empirical performance on short sequence classification tasks. Both directions extend naturally from the sequential modeling philosophy established by the LSTM primary model. Beyond architecture improvements, fine-grained biomechanical scoring — transitioning from binary good/bad classification toward a continuous quality score — would provide richer feedback for coaching applications and better align with how human coaches assess movement quality in practice.

D. Long-Term Vision

The long-term vision for PoseInsight is to evolve from a movement classification system into a fully personalized biomechanical intelligence platform. Integration of large language models, such as Gemini or GPT-4, would enable the system to generate natural language coaching feedback tailored to individual movement errors, replacing generic class labels with actionable, personalized corrective cues. Extension into clinical rehabilitation contexts — where movement quality assessment is critical for post-surgical recovery and injury prevention — represents a high-impact application domain that would benefit from PoseInsight's accessible, camera-based approach. Ultimately, the integration of injury-risk forecasting models that predict musculoskeletal injury probability from detected movement compensations could transform PoseInsight from a reactive coaching tool into a proactive injury prevention system.

XI. Conclusion

This paper presented PoseInsight, a hybrid AI system for real-time exercise form classification and movement quality assessment. By integrating MediaPipe-based pose estimation, biomechanical feature engineering, comparative machine learning evaluation, and interactive Streamlit deployment, PoseInsight evolved from a rule-based prototype into a fully functional intelligent movement analysis framework spanning the complete AI lifecycle from perception to deployment.

The Long Short-Term Memory (LSTM) network was selected as PoseInsight's primary deployed model for its principled alignment with the sequential temporal structure of human movement data, achieving 88% validation accuracy. Exercise quality is inherently a temporal phenomenon — determined by how a movement unfolds across time rather than by any single-frame snapshot — and LSTM's recurrent architecture is specifically designed to capture such sequential dependencies [9]. A comprehensive comparative evaluation additionally demonstrated that Random Forest achieved the highest raw accuracy of 97% on the structured tabular biomechanical feature space, outperforming MLP (85%) and LSTM (88%) under the constrained 168-sample dataset conditions. This result highlights the power of ensemble methods as baselines in low-data regimes and underscores the discriminative

quality of the engineered biomechanical features, while the performance gap relative to LSTM is expected to close as dataset scale increases.

PoseInsight makes several concrete contributions to the field of AI-powered movement analysis. It demonstrates that consumer-grade webcam input combined with interpretable biomechanical features can support reliable exercise quality classification without laboratory equipment or wearable sensors. It establishes a replicable comparative evaluation framework for assessing classical and deep learning approaches under matched conditions. And it provides a deployed, user-facing application that bridges the gap between research-grade model outputs and practical end-user interaction, with LSTM-based sequential inference serving as the production inference backbone.

Looking ahead, PoseInsight establishes a strong foundation for future intelligent coaching systems incorporating personalized natural language feedback, expanded exercise libraries, transformer-based temporal modeling, and clinical rehabilitation applications. The system demonstrates that practical AI success in movement intelligence is achievable today with accessible tools, principled feature engineering, deliberate model selection grounded in problem structure, and careful lifecycle design.

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Appendix

Appendix A: Feature Engineering Implementation Details

Table VIII provides a comprehensive mapping of the 240 extracted features to their corresponding MediaPipe landmark indices and biomechanical interpretations. Features f0–f7 correspond to instantaneous joint angles computed from bilateral knee, hip, elbow, and shoulder landmark triplets. Features f8–f15 capture left-right symmetry ratios computed as the ratio of corresponding left and right joint angles, where a value of 1.0 indicates perfect bilateral symmetry. Features f16–f23 encode frame-to-frame angular velocity signatures representing the rate of change of each joint angle, providing motion dynamics information beyond static posture. The remaining features f24–f239 consist of sliding window mean and standard deviation statistics computed over temporal windows of 5 and 10 frames respectively, applied across all base joint angle channels to capture movement consistency and postural stability across time. For the LSTM primary model, these 240 features are assembled into sequences of 10 consecutive frames before being passed to the recurrent network.

TABLE VIII
Feature Group Mapping for 240-Dimensional Feature Vector

Index Range	Feature Group	Count	Source Landmarks
f0–f3	Knee angles	4	Hip, Knee, Ankle
f4–f7	Hip angles	4	Shoulder, Hip, Knee
f8–f11	Elbow angles	4	Shoulder, Elbow, Wrist
f12–f15	Shoulder angles	4	Neck, Shoulder, Elbow
f16–f19	Symmetry ratios	4	Left/Right angle pairs
f20–f23	Angular velocities	4	Frame-to-frame deltas
f24–f239	Window statistics	216	Mean/std over 5–10 frames
Total	All groups	240	Full body landmarks

Appendix B: Reproducibility Notes

All experiments were conducted on a Kaggle notebook environment using Python 3.12 with the following key library versions: scikit-learn [13] for Random Forest and MLP implementation, TensorFlow/Keras for LSTM development, MediaPipe for pose landmark extraction, and pandas and NumPy for data preprocessing. The dataset used in this study is publicly available on Kaggle at the following URL:

https://www.kaggle.com/datasets/satyabratadas10/pose-insignt-dataset

The dataset consists of a single CSV file (`dataset.csv`) containing 168 labeled samples across four movement-quality classes, with 240 biomechanical features per sample extracted from exercise video recordings using MediaPipe Pose. Model training was performed on an NVIDIA Tesla T4 GPU provided by the Kaggle computing environment. All random seeds were fixed at 42 across train-test splits, model initialization, and Random Forest bootstrap sampling to ensure reproducibility of reported results. The complete training notebook and preprocessing pipeline are available alongside the dataset on Kaggle to support full replication of the reported experiments.